



A Regime-Aware Macro Overlay for Equity Risk Allocation

Presented by Brandon Daniels
February 21, 2026 · Revised September 29, 2026



Executive Summary

- Objective - Develop a systematic, macro-aware predictive signal to forecast 5-Day S&P 500 returns and evaluate its potential integration into an overlay allocation strategy.
- **Key Findings (Corrected, Out-of-Sample)**
- Cross-validated IC: **0.088** (all features), **0.100** (fold-safe top 25)
- Walk-forward IC: **0.006** (HGBR), **0.033** (Ridge)
 - Signal does not persist when retrained through time on past data only
- Overlay Sharpe at 10% vol target: **0.63** vs **0.78** for Buy-and-Hold
- **Strategy Implications**
- Overlay does not beat Buy-and-Hold on a risk-adjusted basis
- Lower drawdowns (-16.9% vs -22.9%) and tail risk, mainly from holding less equity
- Original results (Sharpe 2.11) reflected look-ahead bias, now corrected

What Changed After Review

- **Three look-ahead issues were found and corrected:**
 - Features were chosen using test-period data → now chosen from training data only
 - Volatility target was mis-scaled (ran at ~5%) → now uses daily volatility
 - Training labels overlapped the test week → 5-day purge gap added

Metric	Original method	Corrected
Walk-forward IC (HGBR)	0.138	0.006
Overlay Sharpe (10% vol target)	1.89	0.63
Buy & Hold Sharpe (10% vol target)	0.69	0.78
Overlay max drawdown	-7.4%	-16.9%
Overlay CAGR	7.4%	5.2%

Both columns use the same data (Jan 2005 – Sep 29, 2026). The February deck reported a Sharpe of 2.11 on data through Feb 20, 2026.

I. Research Framework



Research Question

- Can cross-asset macro signals improve 5-day SPY (S&P500) timing?
 - Focus: regime identification and probabilistic ranking
 - Goal: stable OOS IC + implementable Sharpe improvement



Data & Feature Architecture

- Market Indices:
- SPY (SP500),
- DXY (Dollar Index) ,
- VIX (Volatility),
- VIX9D (Volatility 9 Day),
- IXIC (Nasdaq),
- DJI (Dow Jones),
- ES (SP500 Futures),
- NQ (Nasdaq Futures),
- TNX (10 Year Yield),
- FVX (5 Year Yield),
- IRX (13 Week Yield),
- GC (Gold),
- CL (Oil),
- N225 (Nikkei),
- MSCI (International)

Macro: GDP, CPI (FRED), lagged 1 quarter / 1 month

70 engineered features

Top 25 re-selected yearly using past data only

Data window: Jan 2005 – Sep 29, 2026



Modeling Framework

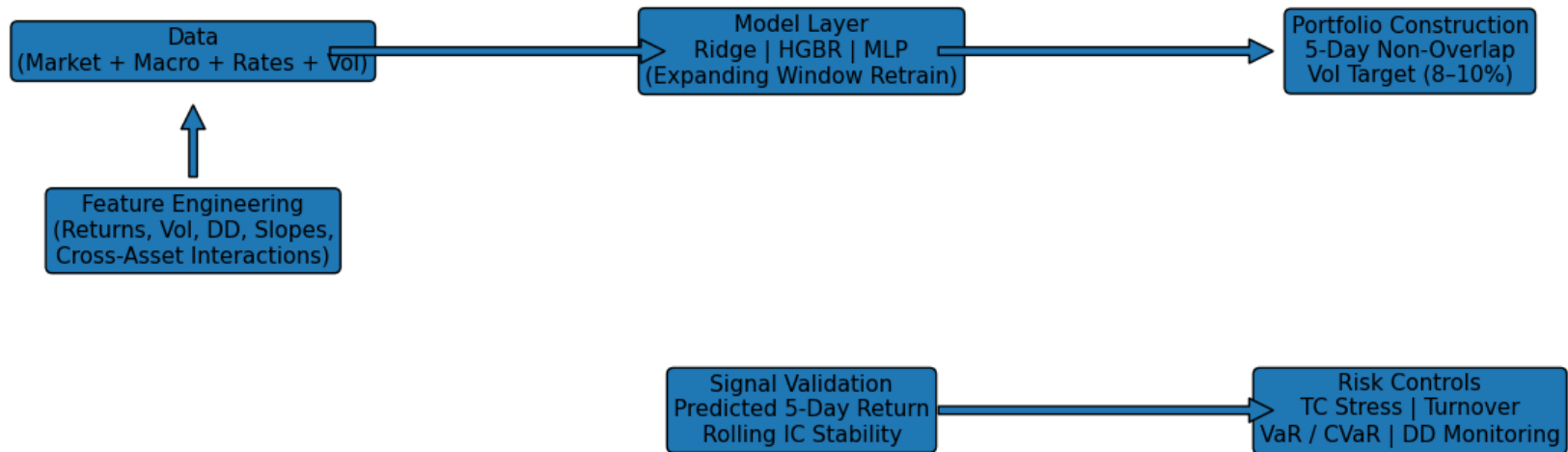
- Ridge Regression (linear benchmark)
- HistGradientBoostingRegressor (nonlinear macro interactions)
- Multi-Layer Perceptron (robustness test)
- Ensemble (70% HGBR / 30% Ridge)

- Deployment Methodology:
 - Expanding-window retraining (live simulation)
 - 5-day non-overlapping decision framework
 - Volatility targeting (8–10%) from daily returns
 - Transaction cost stress testing
 - 5-day purge gap between training and test data
 - Features re-selected yearly from past data only

- Evaluation Metrics:
 - Spearman IC (cross-sectional predictive power)
 - Rolling IC persistence
 - Decile separation
 - Overlay Sharpe
 - Max drawdown
 - VaR / CVaR

Modeling Framework Continued

Systematic Macro Overlay — Modeling Architecture

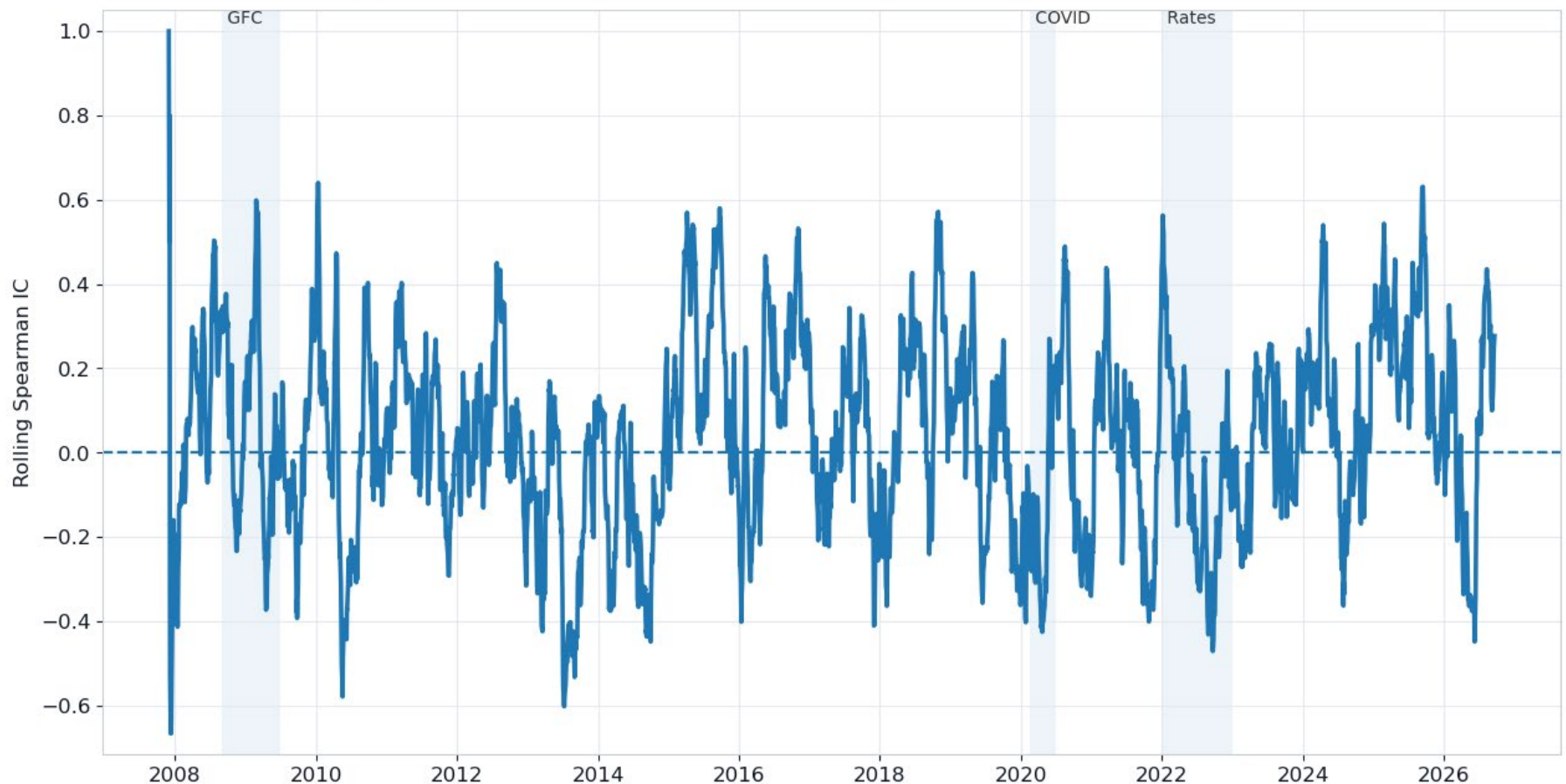


II. Signal Validation

Predictive Stability

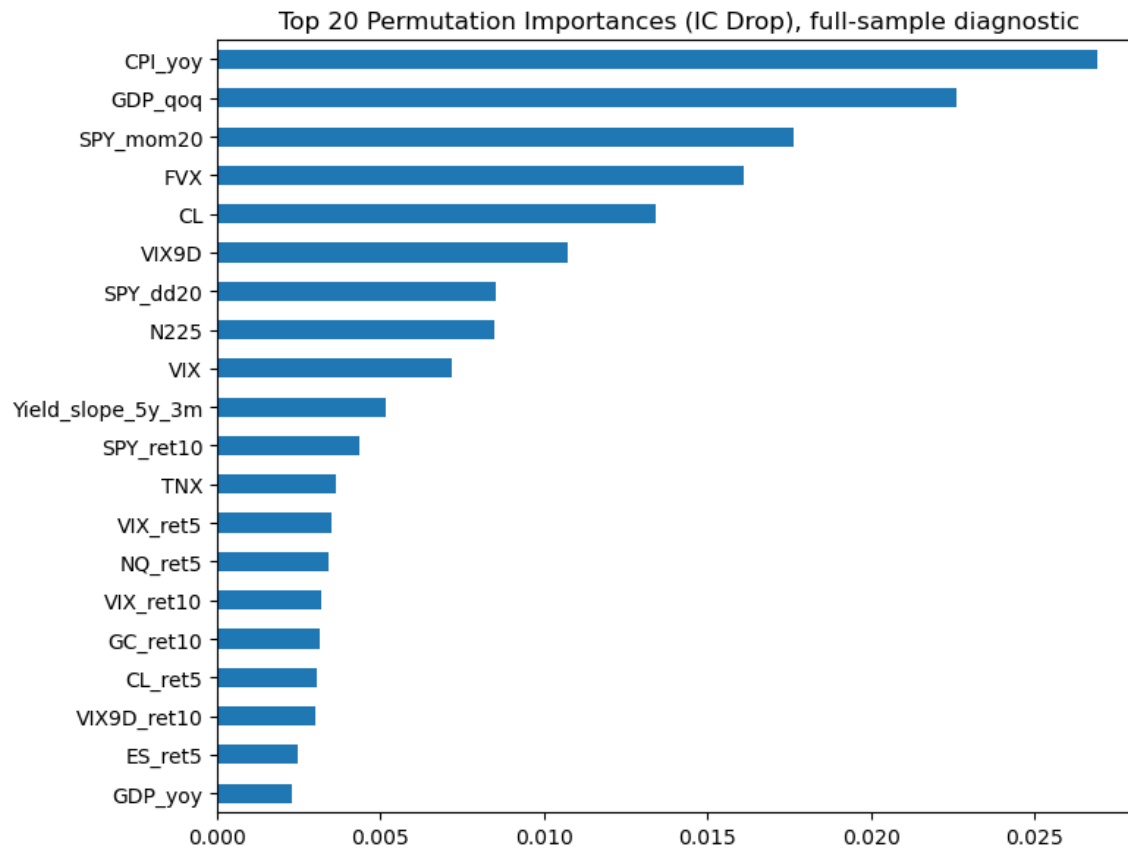
- Cross-validated IC: **0.088** (all features) / **0.100** (fold-safe top 25)
 - Walk-forward IC: **0.006** (HGBR), **0.033** (Ridge); rolling IC swings around zero

Rolling 12-Month Spearman IC — Expanding Window OOS (HGBR)



5-day non-overlapping decisions | Expanding window retrain | Rolling window $\approx$ 12 months (52 obs)

Top Structural Drivers



- **Full-sample ranking**
 - Diagnostic only; not used to train the models
- **Most selected in walk-forward**
 - CPI YoY (100% of retrains)
 - VIX 5-day change (100%)
 - Dow level (90%)
 - SPY 20-day momentum (90%)
 - SPY 20-day drawdown (79%)
 - GDP YoY (74%)

III. Portfolio Construction

Overlay Implementation

- Non-overlapping 5-day holds
 - Long SPY when predicted return $>$ threshold, otherwise cash
 - 5 bps transaction cost per position change

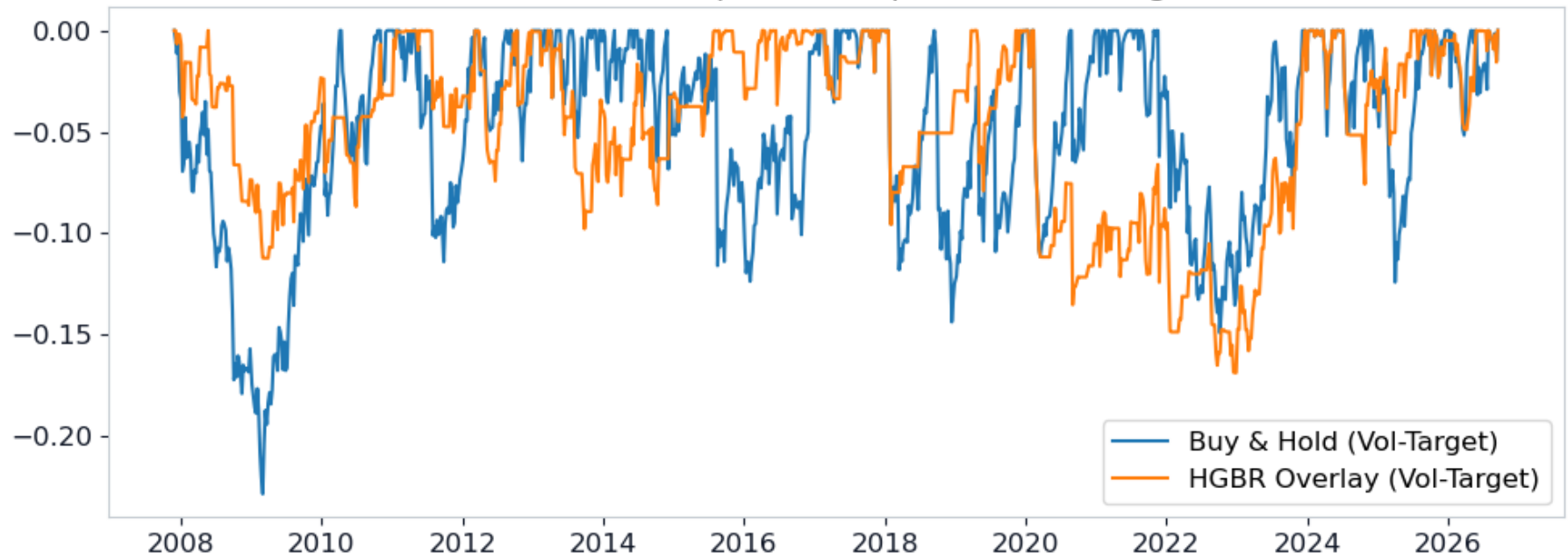
Equity Curves: Best Overlay (Non-overlapping 5-day holds)



Equal 10% Volatility Comparison

- Daily-return volatility scaling (10% target)
 - Overlay Sharpe 0.63 vs 0.78 for Buy-and-Hold
 - Max drawdown -16.9% vs -22.9%, at lower realized vol (8.3% vs 11.3%)

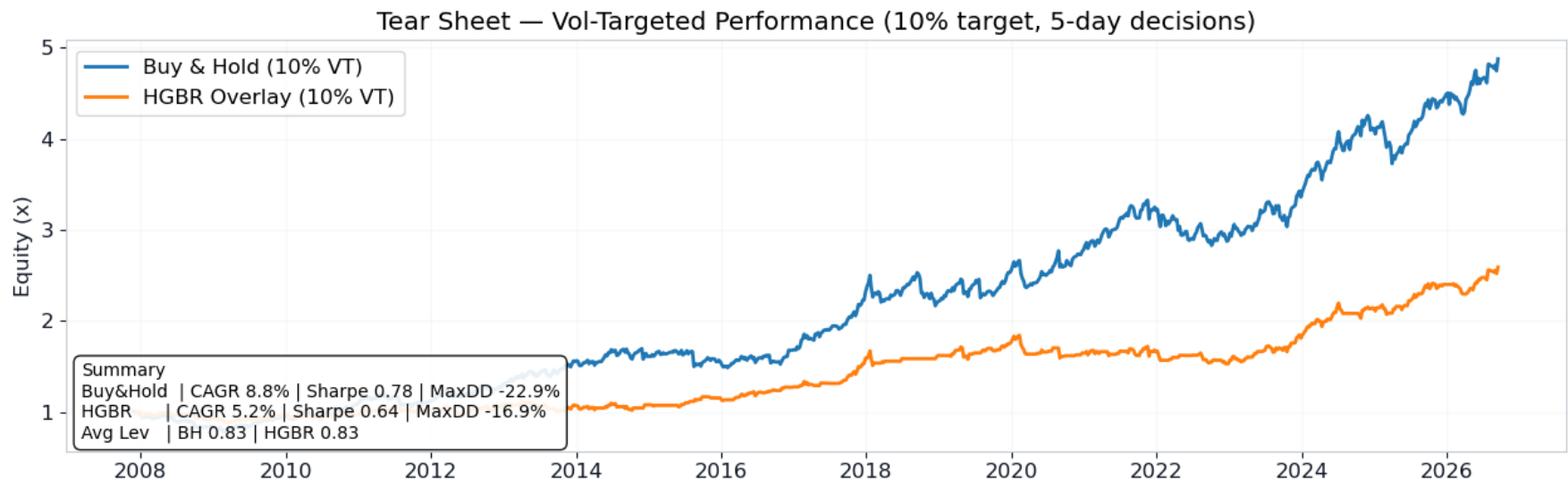
Drawdown Comparison (Equal 10% Vol Target)



IV. Institutional Diagnostics

Equal-Risk Performance (10% Target)

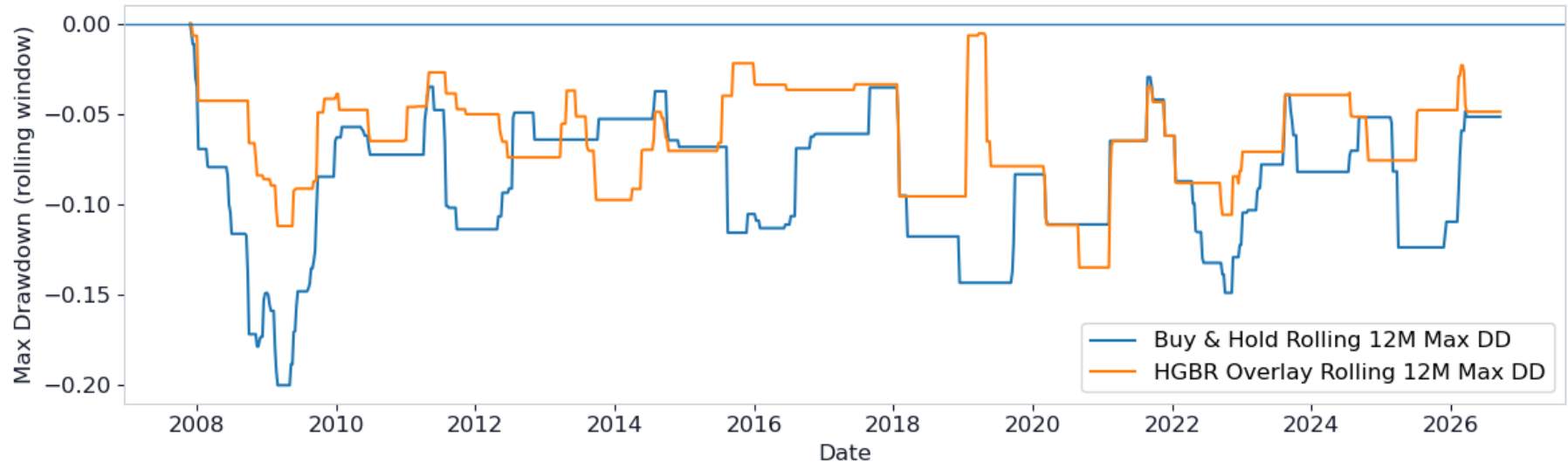
- Sharpe: 0.78 (Buy-and-Hold) vs 0.63 (Overlay)
 - CAGR: 8.8% vs 5.2%
 - Max drawdown: -22.9% vs -16.9%



Rolling 12M Max Drawdown

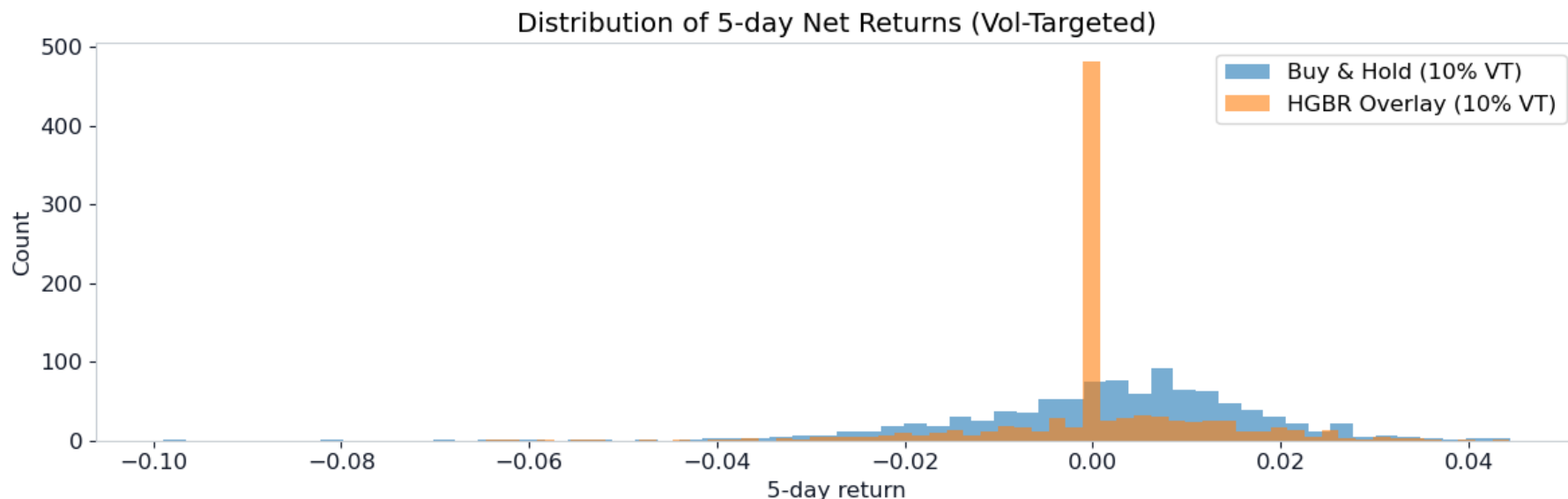
- Overlay drawdowns are usually shallower, especially in 2008–09
 - Largely reflects lower equity exposure, not better timing

Rolling ~12-Month Maximum Drawdown (5-day cadence)



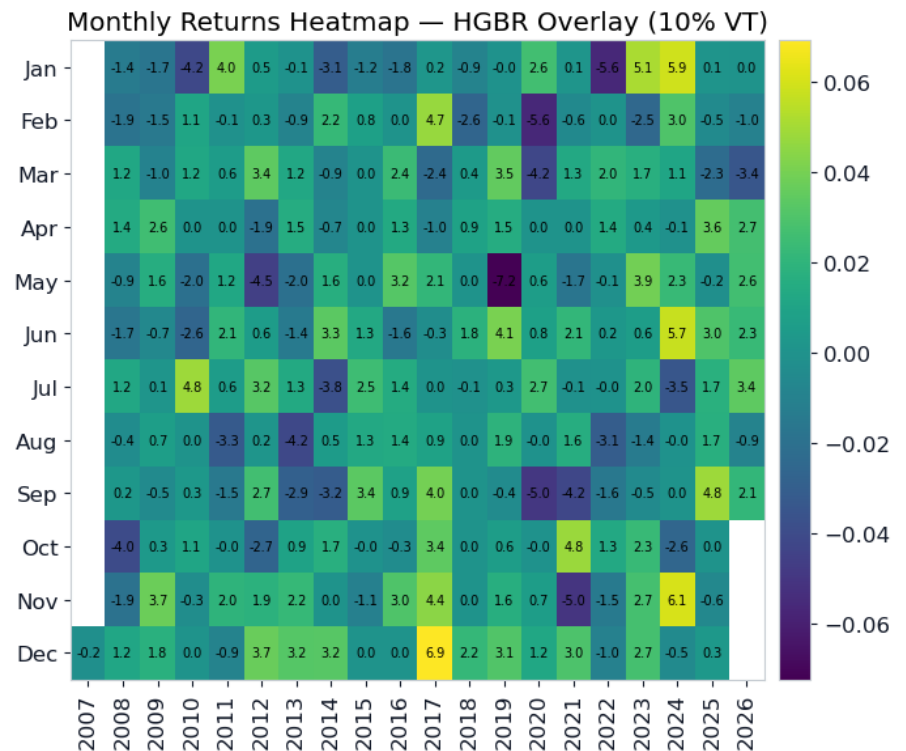
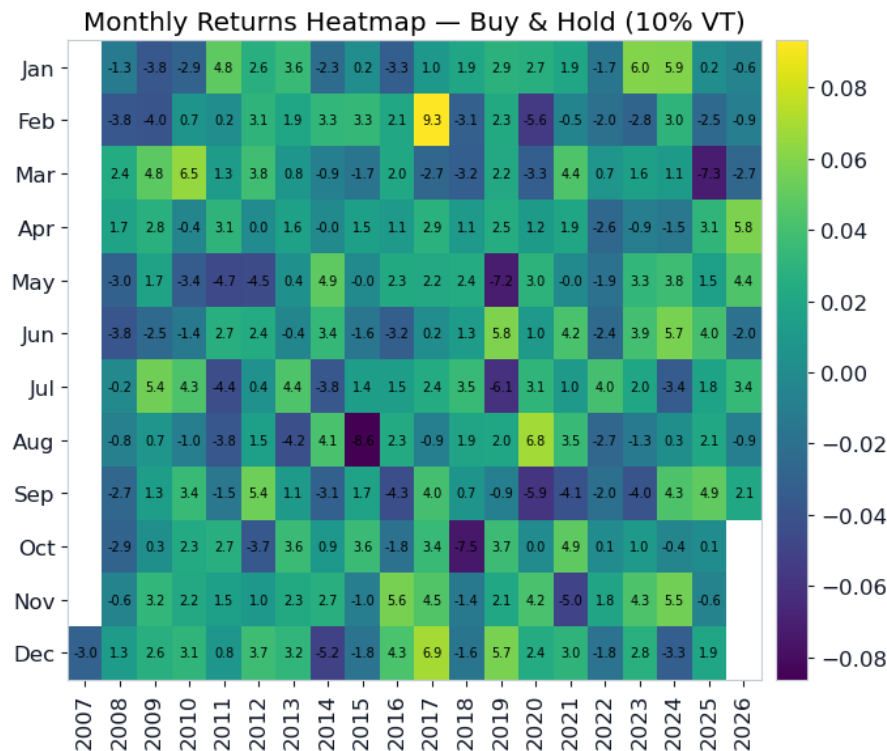
5-Day Return Distribution & Tail Risk

- 5-day VaR (5%): -2.5% $\rightarrow$ -1.9% (~24% lower)
 - CVaR (5%): -3.9% $\rightarrow$ -3.0% (~23% lower)
 - Spike at 0% = weeks in cash; lower exposure drives the reduction



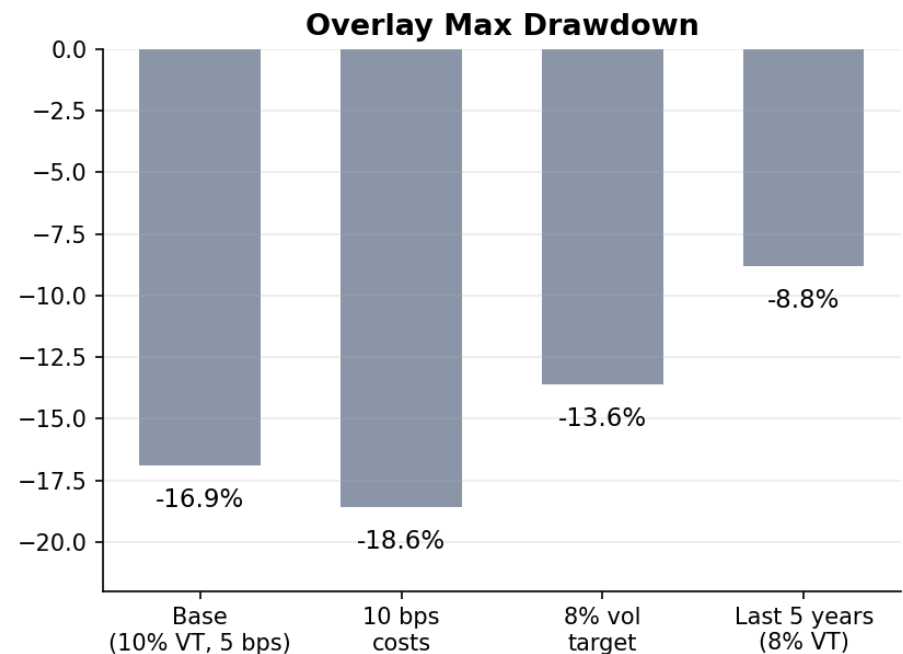
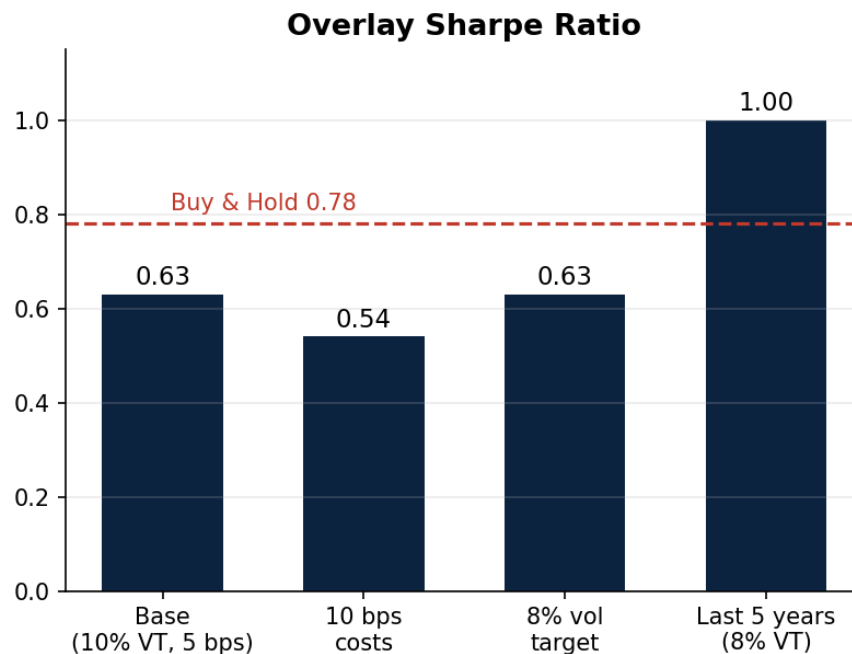
Monthly Return Heatmap

- Positive months: 54% (Overlay) vs 63% (Buy-and-Hold)
 - Worst month -7.2% vs -8.6%; average month 0.45% vs 0.75%



Robustness & Stress Testing

- Sensitive to trading costs: Sharpe 0.63 → 0.54 at 10 bps
 - Sharpe unchanged at 8% vol target (0.63)
 - Last 5 years stronger (Sharpe 1.0), but a short sample
 - Turnover ~0.36 per 5-day period



V. Investment Recommendation



Final Investment Recommendation

- **Not recommended for deployment in its current form**
 - Signal does not persist in walk-forward testing ($IC \approx 0$)
 - Underperforms Buy-and-Hold at equal risk (Sharpe 0.63 vs 0.78)
 - Drawdown reduction comes mainly from lower exposure
- **Next steps**
 - Point-in-time (unrevised) macro data
 - Test other horizons, assets and regime definitions
 - Require a longer untouched holdout before any allocation

Appendix – Full Tear Sheet

Full Tear Sheet Summary

Tear Sheet — Vol-Targeted Performance (10% target, 5-day decisions)

